

A. Row GCD

time limit per test: 2 seconds
memory limit per test: 512 megabytes

You are given two positive integer sequences $a_1, \dots, a_n$ and $b_1, \dots, b_m$. For each $j = 1, \dots, m$ find the greatest common divisor of $a_1 + b_j, \dots, a_n + b_j$.

Input

The first line contains two integers n and m ($1 \leq n, m \leq 2 \cdot 10^5$).

The second line contains n integers $a_1, \dots, a_n$ ($1 \leq a_i \leq 10^{18}$).

The third line contains m integers $b_1, \dots, b_m$ ($1 \leq b_j \leq 10^{18}$).

Output

Print m integers. The j -th of them should be equal to $\text{GCD}(a_1 + b_j, \dots, a_n + b_j)$.

Example

input	output
4 4 1 25 121 169 1 2 7 23	2 3 8 24